

# Emilio Ceballos

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## EDUCATION

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|--------------------------------------------------------------------------------------|-----------------|
| <b>Master of Engineering, Materials Science and Engineering</b> , Cornell University | <b>Dec 2026</b> |
| <b>Bachelor of Science, Mechanical Engineering</b> , Cornell University              | <b>May 2026</b> |

## PROFESSIONAL EXPERIENCE

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| <b>Manufacturing Engineering Intern</b> | <b>Jun — Aug 2026</b> |
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Caterpillar Inc | Brooklyn Park, MN

- Designed iterations for boss-locating fixture, manufacturing it via 3D printing, and testing within manufacturing space
- Developed standard work for fixture preventative maintenance, improving safety and quality scores during audits
- Collaborated with welders to update standard work, ensuring future trainees have the appropriate instructions
- Implemented usage of “tackle boxes” across different work centers in fabrication space, ensuring welders collect and weld the appropriate hardware, minimizing errors

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|-----------------------------------------|-----------------------|
| <b>Manufacturing Engineering Intern</b> | <b>May — Aug 2025</b> |
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Caterpillar Inc | Sanford, NC

- Designed and implemented a vertical holding station for CTL towers, freeing up Automated Guided Vehicles, optimizing flow, and reducing injury risk for operators
- Led Lean Manufacturing cross-functional effort to reduce scraps to Compact Track Loaders (CTL), implementing countermeasures among manufacturing and quality engineers as well as operators
- Developed frame stop within fixture, ensuring part is immobile during welding, improving fixture safety FMEA rating.

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| <b>Advanced Module Engineering Intern</b> | <b>Jun — Aug 2024</b> |
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GlobalFoundries | Malta, NY

- Collected chemical expenditure data across the Cleans module and compared it to theoretical recipe values to ensure tools were properly calibrated
- Developed system using Excel to generate pre-dispense expenditure data and quantify chemical consumption

## PROJECTS

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|-------------------------------------------------------------------------------------|---------------------------|
| <b>CFD Analysis of Geometry and Boundary Conditions on Mixed-Cell Raceway (MCR)</b> | <b>Jan 2026 — Present</b> |
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- Developed Computational Fluid Dynamics model to understand how different velocity magnitudes and directions along with inlet jet and drain geometries influenced particle behavior within novel aquaculture system
- Implemented Discrete Phase Model (DPM) and User-Defined Functions (UDF) to model physics of fish swimming, measuring impact on particle survival time and system efficiency

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| <b>Chassis Subsystem Lead, Cornell Electric Vehicles</b> | <b>Oct 2022 — May 2025</b> |
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- Led a 7-member team in the design and manufacturing of the 2025 Urban Concept vehicle, overseeing carbon-fiber chassis development and structural component production using vacuum infusion and wet layup processes
- Conducted FEA structural analysis in Ansys and optimized composite layup schedules, reducing chassis weight by 10% and lowering material costs by over \$500 while maintaining structural integrity
- Improved composite manufacturing operations by streamlining mold preparation and pre-layup processes, cutting lead times by one week per part, and managed team-wide procurement including sourcing, purchasing, and quality inspection

## CAMPUS INVOLVEMENT

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| <b>Co-President, Society of Hispanic Professional Engineers (SHPE)</b> | <b>Jan 2025 — May 2026</b> |
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- Launched massive recruitment strategy, including a social media overhaul, the creation of a newsletter, and a significant increase in number of events hosted, resulting in a 50% membership retention increase
- Managed coordination, financials, and preparation events to take 22 people to the 2025 SHPE National Convention

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| <b>Racket Service Technician, Cornell Tennis</b> | <b>Aug 2023 — Present</b> |
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- Collaborated with small team to string 30+ rackets weekly under pressing and changing timelines for Cornell's D1 team

**Honors:** Hispanic Scholarship Fund (HSF) Scholar (2022), NACME Corporate Scholar (2024), HSF STEM Summit Scholar (2025)

**Affiliations:** SHPE at Cornell (Academic Chair), M.E.Ch.A (Treasurer), Cornell MEng. Ambassador

## SPECIALIZED SKILLS

**Programs:** Autodesk Inventor, Creo Parametric, Fusion360, Ansys Mechanical, Ansys Fluent, MATLAB, Microsoft Suite

**Languages:** English (Native), Spanish (Native)